Servo Control Software

ESC204, Engineering Science Praxis III
University of Toronto

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Overview

The servo deployment mechanism is controlled using MicroPython running on the Raspberry Pi Pico. The control code, shown in Figure 1, is adapted from the `servo.py` reference provided in Lab 4 [1] and targets the electrical circuit design described in that same lab. It drives each servo by writing a PWM duty cycle to the assigned GPIO pin, mapping a target angle to the corresponding 1 ms to 2 ms pulse width at 50 Hz.

For the deployment system, this code is applied to two servo instances on separate GPIO pins: one for the MG996R linear actuation servo and one for the DF9GMS latch key servo. No changes to the underlying PWM logic are required since both servos use the same standard control protocol; only the pin assignments and target angles differ between the two.

Listing 5: Servomotor output code

```python
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#
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"""CircuitPython Essentials Servo standard servo example"""
import time
import board
import pwmio
from adafruit_motor import servo

# create a PWMOut object on Pin A2.
pwm = pwmio.PWMOut(board.GP14, duty_cycle=2 ** 15, frequency=50)

# Create a servo object, my_servo.
my_servo = servo.Servo(pwm)
print("Servo shaft is turning now.")

while True:
    for angle in range(0, 180, 5):  # 0 - 180 degrees, 5 degrees at a
        time.
        my_servo.angle = angle
        time.sleep(0.05)
    for angle in range(180, 0, -5): # 180 - 0 degrees, 5 degrees at a
        time.
        my_servo.angle = angle
        time.sleep(0.05)
```

Figure 1: Servo control code adapted from the Lab 4 `servo.py` reference [1].

References

[1] University of Toronto Engineering Science, *ESC204S Winter 2026 Lab 4 Instructions*, Version 1.0, January 29, 2026.